

Output Ontology (OO)

Score: 4/5 — Minor gaps | Confidence: high

Benchmark: MILU | Context: North Indian Hindi-Medium Postgraduate Teacher — MCQ Evaluation Deployment

Definition: Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

Justification

Output ontology aligns very well with deployment. MILU produces a single correct option per MCQ drawn from exam answer keys [Q12, Q26], scored as binary correct/incorrect — structurally identical to the deployment's positive/negative marking requirement. The user explicitly confirmed (Q4 elicitation) that no rubric-based or graded output is needed. Domain-level reporting is granular [Q42, Q98]. Two concerns reduce the score from 5: (a) a malformed item where an instruction line ('use the code below') appears as an answer option, indicating output-label corruption [DATASET-D49]; (b) culturally variable correct-answer norms are not discussed in the paper, though the user confirmed evaluation schemas are broadly consistent across boards and competitive exams, mitigating this concern.

ID	Evidence
D49	पंचायतों के संबंध में निम्नलिखित में से कौन सा/से सही है/हैं?...नीचे दिए गए कोड का उपयोग करके सही उत्तर चुनें: — Option4 of the MCQ is an instruction ("use the code below"), not an answer — malformed item

Strengths

- Single-correct-answer label schema matches deployment's binary scoring schema exactly [Q22, Q42].
- Per-domain, per-language, per-subject evaluation tables enable Hindi-specific analysis [Q98, Q99-Q132].
- Culturally relevant domain categorisation (Arts & Humanities, Law & Governance) explicitly reported [Q20, Q71].

Checklist

Checklist item definitions:

ID	Assessment Question
OO-1	Review output label categories for regional relevance
OO-2	Identify missing regionally specific categories
OO-3	Flag categories encoding non-regional values
OO-4	Consider stakeholder-driven taxonomy redesign
OO-5	Document taxonomy issues harming structural/content validity

Checklist responses:

OO-1: Output labels (option1-4 with single target) are regionally relevant — they replicate the standard Indian MCQ answer-key format used by all North Indian boards [Q12, Q22].

OO-2: No regionally specific output category is missing for the binary MCQ scoring use case. Rubric-based or graded output not required per deployment elicitation.

OO-3: INSUFFICIENT DOCUMENTATION on whether any items encode non-regional values; the user noted competitive-exam keys are broadly consistent with North Indian teacher norms (elicitation Q4), but the paper does not analyse culturally variable correct-answer norms.

OO-4: Stakeholder taxonomy redesign not required — schema is a precise structural match.

OO-5: Minor structural-validity concern from a malformed answer option [DATASET-D49] indicating data-quality erosion of the output schema; the issue is item-level rather than schema-level.

ID	Evidence
D49	पंचायतों के संबंध में नमिनलखिति में से कौन सा/से सही है/हैं?...नीचे दिए गए कोड का उपयोग करके सही उत्तर चुनें: — Option4 of the MCQ is an instruction ("use the code below"), not an answer — malformed item

Information Gaps

- Frequency of malformed-option items beyond the validation sample observed.
- Whether any Hindi items have answer keys that diverge from North Indian teacher norms (no documented IAA).

Requires Expert Verification

- Spot-check of Hindi answer keys against UP/MP/Rajasthan/Bihar board model-answer expectations.

Remediation

Gap 1: At least one item observed where a formatting instruction appears as an answer option (DATASET-D49), indicating data-quality erosion of the output label schema.

Recommendation: Run a heuristic audit of all Hindi options to detect instruction-like strings (e.g., 'नीचे दिए गए कोड का उपयोग करके') appearing as answer choices and exclude or correct affected items.